```
Dataset Shape: (19237, 18)

Column Names:
Index(['ID', 'Price', 'Levy', 'Manufacturer', 'Model', 'Prod. year',
       'Category', 'Leather interior', 'Fuel type', 'Engine volume', 'Mileage',
       'Cylinders', 'Gear box type', 'Drive wheels', 'Doors', 'Wheel', 'Color',
       'Airbags'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19237 entries, 0 to 19236
Data columns (total 18 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                19237 non-null  int64
 1   Price             19237 non-null  int64
 2   Levy              19237 non-null  object
 3   Manufacturer      19237 non-null  object
 4   Model             19237 non-null  object
 5   Prod. year        19237 non-null  int64
 6   Category          19237 non-null  object
 7   Leather interior  19237 non-null  object
 8   Fuel type         19237 non-null  object
 9   Engine volume     19237 non-null  object
 10  Mileage           19237 non-null  object
 11  Cylinders         19237 non-null  float64
 12  Gear box type     19237 non-null  object
 13  Drive wheels      19237 non-null  object
 14  Doors             19237 non-null  object
 15  Wheel             19237 non-null  object
 16  Color             19237 non-null  object
 17  Airbags           19237 non-null  int64
dtypes: float64(1), int64(4), object(13)
```

```
dtypes: float64(1), int64(4), object(13)
memory usage: 2.6+ MB

Missing Values:
ID                  0
Price               0
Levy                0
Manufacturer        0
Model               0
Prod. year          0
Category            0
Leather interior    0
Fuel type           0
Engine volume       0
Mileage             0
Cylinders           0
Gear box type       0
Drive wheels        0
Doors               0
Wheel               0
Color               0
Airbags             0
dtype: int64
```

First 5 Rows:

	ID	Price	Levy	Manufacturer	Model	Prod. year	Category	Leather interior	Fuel type	Engine volume	Mileage	Cylinders	Gear box type	Drive wheels	Doors	Wheel	Color	Airbags
0	45654403	13328	1399	LEXUS	RX 450	2010	Jeep	Yes	Hybrid	3.5	186005 km	6.0	Automatic	4x4	04-May	Left wheel	Silver	12
1	44731507	16621	1018	CHEVROLET	Equinox	2011	Jeep	No	Petrol	3	192000 km	6.0	Tiptronic	4x4	04-May	Left wheel	Black	8
2	45774419	8467	-	HONDA	FIT	2006	Hatchback	No	Petrol	1.3	200000 km	4.0	Variator	Front	04-May	Right-hand drive	Black	2
3	45769185	3607	862	FORD	Escape	2011	Jeep	Yes	Hybrid	2.5	168966 km	4.0	Automatic	4x4	04-May	Left wheel	White	0
4	45809263	11726	446	HONDA	FIT	2014	Hatchback	Yes	Petrol	1.3	91901 km	4.0	Automatic	Front	04-May	Left wheel	Silver	4

Duplicate Rows: 313

Missing Values:
```
ID                0
Price             0
Levy              0
Manufacturer      0
Model             0
Prod. year        0
Category          0
Leather interior  0
Fuel type         0
Engine volume     0
Mileage           0
Cylinders         0
Gear box type     0
Drive wheels      0
Doors             0
Wheel             0
Color             0
Airbags           0
dtype: int64

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
Index: 18924 entries, 0 to 19236
Data columns (total 18 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                18924 non-null  int64
 1   Price             18924 non-null  int64
 2   Levy              18924 non-null  object
 3   Manufacturer      18924 non-null  object
 4   Model             18924 non-null  object
 5   Prod. year        18924 non-null  int64
 6   Category          18924 non-null  object
 7   Leather interior  18924 non-null  object
 8   Fuel type         18924 non-null  object
 9   Engine volume     18924 non-null  object
 10  Mileage           18924 non-null  object
 11  Cylinders         18924 non-null  float64
 12  Gear box type     18924 non-null  object
 13  Drive wheels      18924 non-null  object
 14  Doors             18924 non-null  object
 15  Wheel             18924 non-null  object
 16  Color             18924 non-null  object
 17  Airbags           18924 non-null  int64
dtypes: float64(1), int64(4), object(13)
memory usage: 2.7+ MB

Data cleaning completed successfully!
```

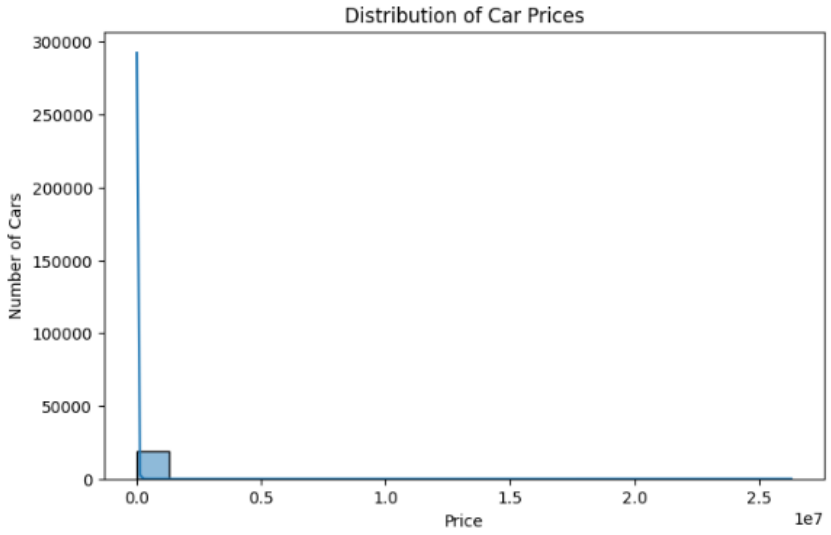
```
Column Names:
['ID', 'Price', 'Levy', 'Manufacturer', 'Model', 'Prod. year', 'Category', 'Leather interior', 'Fuel type', 'Engine volume', 'Mileage', 'Cylinders', 'Gear box type', 'Drive wheels', 'Doors', 'Wheel

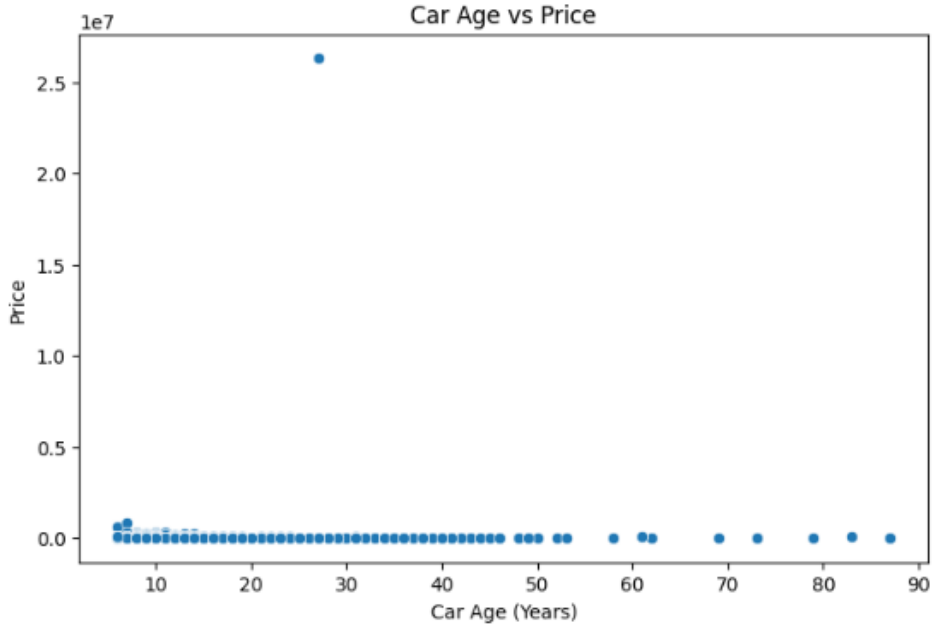
Statistical Summary:
```

	ID	Price	Prod. year	Cylinders	Airbags
count	1.892400e+04	1.892400e+04	18924.000000	18924.000000	18924.000000
mean	4.557538e+07	1.858744e+04	2010.914236	4.580216	6.568379
std	9.375468e+05	1.921356e+05	5.665749	1.200223	4.322323
min	2.074688e+07	1.000000e+00	1939.000000	1.000000	0.000000
25%	4.569501e+07	5.331000e+03	2009.000000	4.000000	4.000000
50%	4.577191e+07	1.317200e+04	2012.000000	4.000000	6.000000
75%	4.580174e+07	2.206300e+04	2015.000000	4.000000	12.000000
max	4.581665e+07	2.630750e+07	2020.000000	16.000000	16.000000

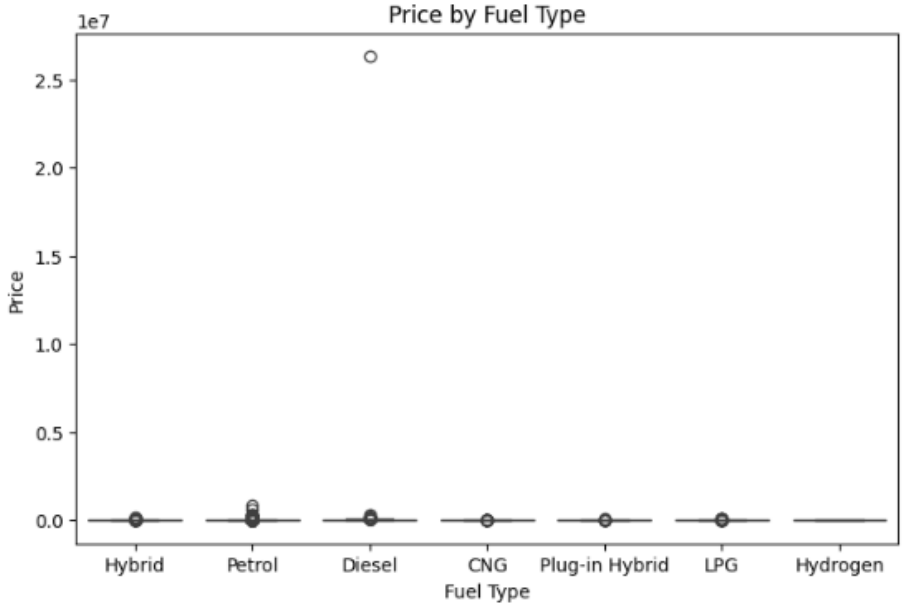
```
Number of Rows: 18924
Number of Columns: 18
```

```
Average Price: 18587.435267385332
Minimum Price: 1
Maximum Price: 26307500
```





```
Loading and cleaning df...
```




```
Features prepared successfully!
Reduced features shape: (18924, 26)
Target shape: (18924,)
```

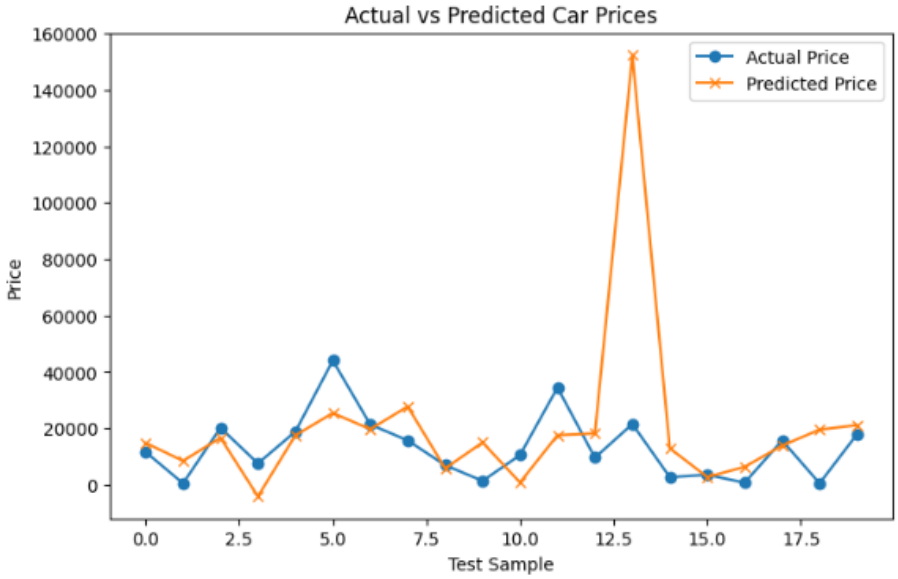
```
Rebuilding X and y datasets...
Training data shape: (15139, 8457)
Testing data shape: (3785, 8457)
Data splitting completed successfully!
```

```
Fitting the model on optimized features...
Model training completed successfully!
```

```
Model Evaluation Results
------------------------
Mean Absolute Error (MAE): 12148.07
Mean Squared Error (MSE): 495494528.57
Root Mean Squared Error (RMSE): 22259.71
R2 Score: -0.4311
R2 Score (%): -43.11 %
```

```
Predicted Car Price: 14805.96
Actual Car Price: 11604
```

...


Actual vs Predicted Car Prices
Actual Price
Predicted Price
Price
Test Sample
160000
140000
120000
100000
80000
60000
40000
20000
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5